

DAS/Detector shown at 12 o'clock
Looking from the Table side of the Gantry



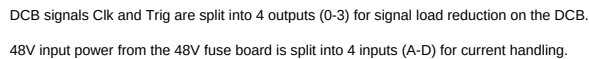
CLK (from DCB)

TRIG (from DCB)

48V POWER
(from fuse board)

24VDC

1 2 3 4



The diagram illustrates the mapping of physical diode data to a scan data format. On the left, a box labeled "Physical Diode" is divided into two vertical sections: "Side 1" on the left and "Side 2" on the right. A large white arrow points from this box to a box on the right labeled "Scan data format".

The "Scan data format" box is divided into two horizontal sections. The top section is labeled "B side" and contains rows "Row 32B" at the top and "Row 1B" at the bottom. The bottom section is labeled "A side" and contains rows "Row 1A" at the top and "Row 32A" at the bottom. A horizontal dotted line connects "Row 1B" in the "B side" section to "Row 1A" in the "A side" section, indicating a mapping between the two sides.

Signals are translated back into the same format as the VCT Sherlock (VDAS) detector before they are sent to the host so scan data looks the same for all VCT detectors.

Special information for decoding firmware addressing if seen in error messages:

Errors on the FRDMs are reported by the DIFBs. FRDMs are also referred to as Digital Modules. ASIC errors will be reported by DIFB # and ASIC number. As each DIFB handles 8 A/D cards, it can report errors for FRDM ASICs 0 - 63. This is decoded by firmware addressing of the A/D cards.

Reported ASIC #	FRDM FW Address
0 - 15	0
16 - 31	1
32 - 47	2
48 - 63	3

VCT HD DAS Detector Block Diagram Rev 1.xls